Surface Tension of Liquids

What is Surface Tension?

- Surface tension is the phenomenon that occurs when the surface of a liquid is in contact with another phase (it can be a liquid as well).
- Liquids tend to acquire the least surface area possible.
- The surface of the liquid behaves like an elastic sheet.

"Surface tension is the tension of the surface film of a liquid caused by the attraction of the particles in the surface layer by the bulk of the liquid, which tends to minimise surface area".

- Surface tension not only depends upon the forces of attraction between the particles within the given liquid but also on the forces of attraction of solid, liquid or gas in contact with it.
- The energy responsible for the phenomenon of surface tension may be thought of as approximately equivalent to the work or energy required to remove the surface layer of molecules in a unit area. Surface tension may be expressed, therefore, in units of energy (joules) per unit area (square metres).
- Surface tension is typically measured in dynes/cm, the force in dynes is required to break a film of length 1 cm.

Surface tension of various liquids:

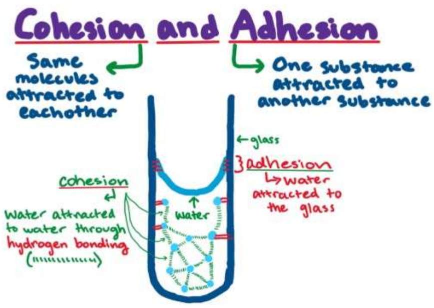
Liquid	Surface Tension (N/m)
Hydrogen	2.4
Helium	0.16
Water	0.072
Ethanol	22.0
Sodium Chloride	114

Cohesive forces

The surface tension of a liquid results from an imbalance of intermolecular attractive forces, the cohesive forces between molecules. The molecules at the surface of this sample of liquid water are not surrounded by other water molecules. The molecules inside the sample are surrounded by other molecules. A molecule in the bulk liquid experiences cohesive forces with other molecules in all directions. A molecule at the surface of a liquid experiences only net inward cohesive forces.

Adhesive Forces

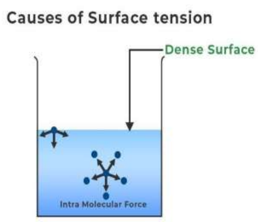
Forces of attraction between a liquid and a solid surface are called adhesive forces. The difference in strength between cohesive forces and adhesive forces determine the behavior of a liquid in contact with a solid surface. Water does not wet waxed surfaces because the cohesive forces within the drops are stronger than the adhesive forces between the drops and the wax. Water wets glass and spreads out on it because the adhesive forces between the liquid and the glass are stronger than the cohesive forces within the water.



What Causes Surface Tension?

The effect known as surface tension is caused by the cohesive forces between liquid molecules. Since the molecules at the surface lack like molecules in both directions, they cohere more closely to those specifically aligned with them on the surface. Intermolecular forces such as Van der Waals force, draw the liquid particles together. Along the surface, the particles are pulled toward the rest of the liquid. This creates a surface "film" that makes moving an object across the surface more difficult than moving it while fully submerged.

Let us assume a jar is filled with water; the water molecules can be found in two positions in this jar: First, beneath the water, and second, on the surface of the water. Since there are no molecules above these molecules, the molecules at the water's surface are unbalanced. As a result, only the molecules below will be attracted. As a result, a thin crust will form on the liquid's upper surface. Because of this thick layer, a form of stress is generated, which is known as Surface Tension. These phenomena can also be explained in terms of energy.



Surface Tension at Molecular Level

Due to the Cohesion force the water molecule tends to stick together. The water molecule at the bottom layer has various molecules above them to stick but the molecule at the top layer does not have various other molecules to cling together. Thus, they attach to each other with a larger force and resist any change in their structure. The molecule inside the body of the liquid experiences the forces from all directions and thus, the net force cancels out each other, whereas the particle at the top layer experiences a strong inward force resulting in the surface tension of the water. Because of this water has one of the highest surface tension among liquids.

Formula for Surface Tension

Mathematically, the surface tension is defined as the force (F) acting on the surface and the length (l) of the surface, so is given as:

$$\mathbf{T = F / l}$$

Also, the ratio of the work done (W) and the change in the area of the surface (A) is termed surface tension.

Unit of Surface Tension

SI Unit	N/m (Newton per Meter)
CGS Unit	dyn/cm

Dimension of Surface Tension

The dimension formula of Force is $[MLT^{-2}]$ whereas the dimension of length is [L] thus, the dimensional formula of **Surface Tension is $[M L^0 T^{-2}]$.**

Some Important Examples of Surface Tension

a. Insects Walking on Water

Various insects can easily walk on the surface of the water because the force of their weight is not enough to penetrate the surface of the water

b. Floating Needle

A needle made of steel can easily be made to float on the surface of the water even though it is many times denser than water because of the surface tension of the water.

c. Rise of Liquids in Capillary Tubes

A tube whose radius is very short and uniform is called a capillary tube. When an open capillary is dipped in water. The water rises to some height in the capillary tube.

d. Spherical Shape of Water Droplets

Small droplets of fluid are spherical due to surface tension. The molecules of water tend to stick together due to intramolecular force, and the energy of molecules that are located on the surface of droplets contains higher energy and try to push the other molecule to the centre of the droplet. Due to this the drop makes the shape that contains the least surface area and the spherical shape is best for the least surface area, that's why The droplets of water and raindrops are spherical.

e. Fire Polishing of Glass

The method of polishing glass or thermoplastic with the help of fire or flame is called Fire Polishing. When we heat a glass material in flames, the glass surface starts melting. But due to surface tension, it starts to become soft and smooth which makes the glass very flat and smooth. This method is most applicable to flat external surfaces. Flame polishing is frequently in acrylic plastic fabrication because of its high speed compared to abrasive methods. In this application, an oxyhydrogen torch is typically used, one reason being that the flame chemistry is unlikely to contaminate the plastic.

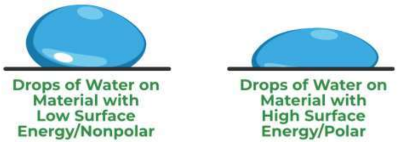
f. Soaps and Detergents

Soaps and Detergents can easily clean clothes because they lower the surface tension of the water and thus allowing it to easily soak the grease and soil particles and then remove them.

What is Surface Energy?

Surface energy measures the breakdown of intermolecular bonds caused by the formation of a surface. Surface-free energy and interfacial-free energy are other names for it. Surface energy is defined as the work done per unit area by the force that forms the new surface. When the free surface area of a liquid is increased, effort must be done against the force of surface tension. This work is stored as potential energy on the liquid surface. This increased potential energy per unit area of the free surface of the liquid is referred to as surface energy.

The image given below shows the surface of the water molecules.



Mathematically, the surface energy is defined as:

$$\textbf{\textit{Surface energy}} = \textbf{\textit{Surface tension}} \times \textbf{\textit{Change in the surface area}}$$

or

$$E_S = T \times \Delta A$$

where T denotes surface tension and ΔA denotes an increase in surface area.

Therefore, the **SI unit of surface energy is Nm^{-2} and the dimensional formula is [MT^{-2}].**

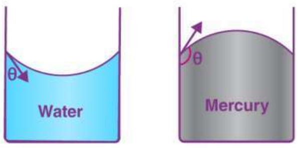
What is Angle of Contact?

The **angle of contact (θ)** is defined as the angle subtended between the tangents drawn at the liquid surface and the solid surface within the liquid at the point of contact.

The angle of contact varies from 0^o to 180^o.

If the adhesive force is greater than the cohesive force, the contact angle is less than 90o. This can be seen between water and glass surfaces.

If the adhesive force is less than the cohesive force, the contact angle is greater than 90o. This can be seen between mercury and glass surfaces.



(A) Acute angle (B) Obtuse angle

The angle of contact depends on the following factors:

- The nature of the liquid, the solid with which it comes into contact.
- The medium that exists above the free surface of the liquid.
- As the temperature of the liquid rises, so does the angle of contact.
- When soluble impurities are added to a liquid, the angle of contact drops.

Factors affecting Surface Tension

Various factors which affect the Surface Tension of any liquid are

- **Solubility:** If the solute is highly soluble in the fluid, the surface tension of the fluid would increase. And if the solute is less soluble in the fluid, then the surface tension of the fluid would decrease.

- **Dust particles and lubricants:** If there are dust particles or any lubricant present on the surface of the fluid, the surface tension of the fluid decreases.
- **Temperature :** Increasing the temperature reduces the surface tension of the fluid. And decreasing the temperature increases the surface tension.

Measurement of surface tension:

Various methods can be used to measure the surface tension of any liquid. For example:

- Capillary Rise Method
- Bubble Pressure Method
- Spinning Drop Method
- Du Noüy–Padday Method
- Du Noüy Ring Method
- Stalagmometric Method
- Sessile Drop Method

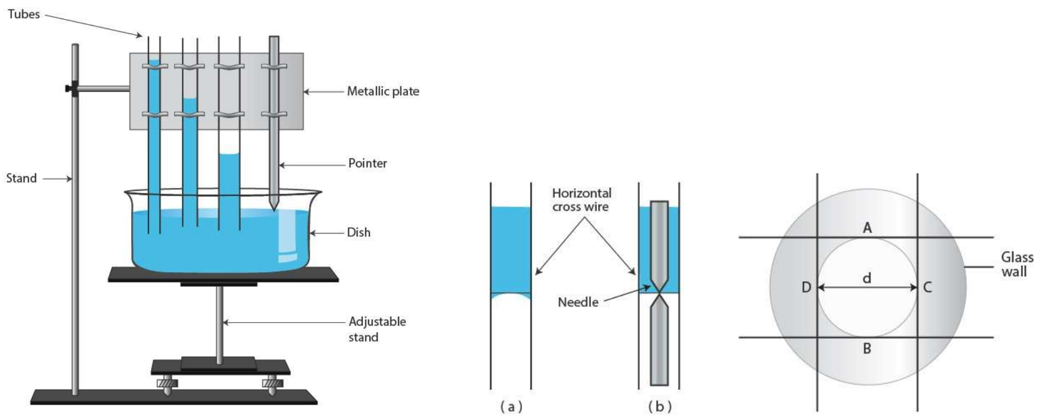
Surface Tension by Capillary Rise Method

When we experiment with a capillary tube, we observe that when a liquid rises in a capillary tube, the weight of the column of the liquid of density ρ inside the tube is supported by the upward force of surface tension acting along the circumference of the points of contact.

A liquid of density ρ and surface tension σ rises in a capillary of inner radius 'r' to a height 'h' is given by:

$$h = 2\sigma \cos\theta/\rho g r$$

where, θ = The contact-angle made by the liquid meniscus with the surface of the capillary.



Measurement of surface tension by capillary rise — Water meniscus through microscope — Measurement of internal diameter of capillary tube

The surface tension of water is given by the formula

$$T = \frac{r(r+h/3)\rho g}{2\cos\theta}$$

where, r is the radius of cross-section, g is the acceleration due to gravity, ρ is the density of the liquid, h is the capillary rise, θ is the contact angle.